Coding & Solution

Project Title: Rising Waters

Introduction

The Rising Waters project is developed using Machine Learning and Flask to predict the possibility of floods based on weather and rainfall parameters. The application provides users with an easy-to-use interface for entering data and receiving flood predictions instantly.

Coding Approach

The project is developed using Python for implementing the Machine Learning model and Flask for creating the backend application. The frontend is designed using HTML, CSS, and JavaScript to provide a simple and responsive user interface. The trained model and scaler are saved using Joblib and loaded automatically when the application starts.

Solution Implementation

The system accepts weather parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and subdivision rainfall from the user. The input data is preprocessed using the saved scaler before being passed to the trained XGBoost model. The model predicts whether flood conditions are likely to occur and calculates the prediction probability. Finally, the application displays either a **Flood Risk** or **No Flood Risk** result along with the prediction probability.

Technologies Used

The project is implemented using Python, Flask, HTML, CSS, JavaScript, Pandas, NumPy, Scikit-learn, XGBoost, and Joblib.

Advantages

- Accurate flood prediction using Machine Learning.
- Fast prediction with probability score.
- User-friendly web interface.
- Easy to maintain and extend.
- Supports disaster preparedness.

Conclusion

The coding solution integrates Machine Learning with a Flask-based web application to provide an efficient flood prediction system. The developed solution offers accurate predictions, quick response time, and a simple interface that helps users identify potential flood risks effectively.